Simultaneous Segmentation and Anatomical Landmark Detection on 3D Data using Context-aware Multi-task Learning

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Abstract

Segmentation and anatomical landmark detection are important tasks needed for digital planning of surgery. The accuracy of the output from these tasks greatly influence the outcome of downstream tasks such as digital templating and implant size selection. Most of the existing methods treat these tasks separately. However, they are inherently inter-related and can benefit when handled together. In this work, we propose a multi-task network which simultaneously segments and detects the anatomical landmarks in 3D images. The proposed global context aware multi-task network allows for excellent predictions even under limited data. We apply our method on 3D CT volumes of pelvis and through experimental results show that each of the tasks benefit from joint learning, thus providing better performance when compared to dedicated network for each task.

Keywords

Multi-task learning, Anatomical landmark detection, Segmentation, 3D pelvis CT.

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1 INTRODUCTION

Pre-operative planning is vital step for any surgery such as total hip arthroplasty and total knee arthroplasty. Pre-operative planning helps to reduce the intra-operative and post-operative complications [1]. Correctly sized components are essential as oversizing can result in reduced joint function and early component failure, whilst undersized implant components may result in instability and recurrent dislocations. The effectiveness of planning and the end results of surgery depends highly on the accuracy of the templating process. This in turn depends on accurate identification of anatomical landmarks and boundaries of various bones.

Anatomical landmarks are distinct point on anatomical structures. Accurate identification of these landmarks are crucial as all the downstream tasks depend on it. Traditionally, anatomical landmarks are identified manually by an expert during treatment planning. This is a tedious time consuming process. The identification of landmarks is crucial on intra-operative scans as well, which makes quick automatic identification extremely important. Identification of anatomical landmarks on 2D images is well studied. In Lindner *et. al.* [7], random forest regression voting along with statistical shape analysis was shown to provide good accuracies on cephalometric radiographs. In Payer *et. al.* [8], the learning task was split into a dedicated network, which learns the locally accurate but ambiguous predictions, whereas a second network filters the ambiguous predictions by learning the global context. In Subburaj *et. al.* [10], automatic identification of anatomical landmarks was performed on a 3D model reconstructed from CT images.Fischer *et. al.* [3] used an iterative tangential plane method for automatic identification of landmarks on surface models of the pelvis.

Precise segmentation of bone from 3D CT scans has been of great interest in the context of orthopedic surgical planning [13]. Some of the earliest work in this includes statistical shape and intensity modeling. In Krčah *et. al.* [6], graph-cuts and the second-order local structure analysis was presented. Recently deep learning based methods have shown good performance when applied on individual 2D slices separately and also on 3D volumes directly [17].

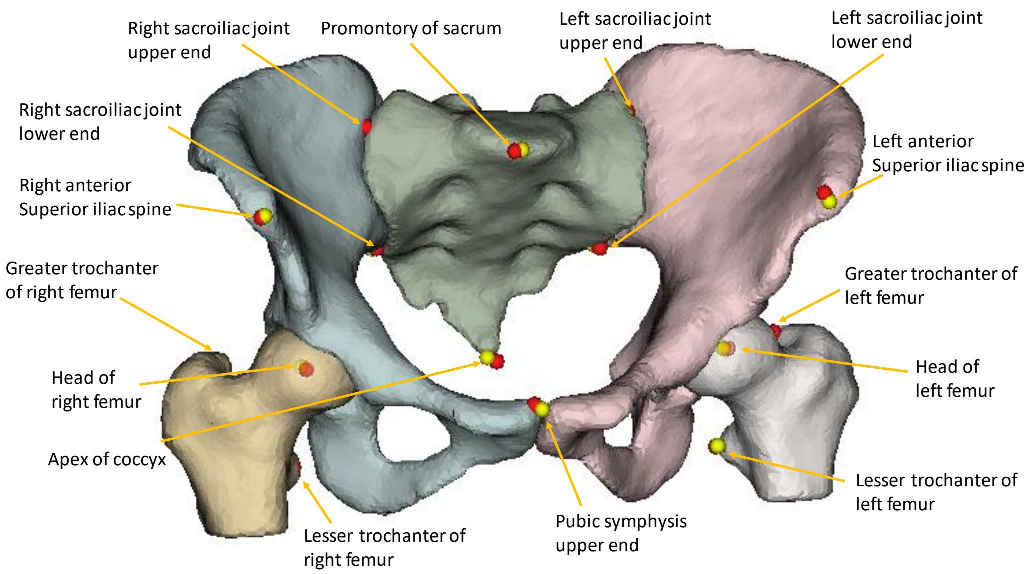


Figure 1: 3D rendering of the predicted segmentation masks using proposed method (Right and left femur, right and left hip, and Sacrum, each highlighted with different color). 15 landmarks used in this study are also shown. Ground truth annotation is indicated by red points and predicted landmarks with yellow.

In the existing work, detection of landmarks and segmentation of bones are treated as two independent tasks. The landmarks of interest are usually located on or within the boundaries of the bones, which makes the task of bone segmentation and landmark detection highly related. Recently, it has been shown that learning multiple related tasks can boost the performance of the network in various applications [15][2] including medical image analysis [16]. By utilizing the relative commonality between the tasks these methods show that the performance of each individual task can be improved. In Zhang *et. al.* [14] a framework for a joint bone segmentation and landmark digitization framework for craniomaxillofacial bone was proposed.

In this work, we present a novel convolutional neural network based method for joint segmentation and landmark detection on 3D volumes. The main contributions presented in this work are: i) we propose a multi-task CNN architecture that uses a global context aware decoder branch for anatomical landmark detection; ii) we use a 3D feature fusion block to share the information between these two closely related tasks; iii) we apply the method on CT volumes of pelvis to predict the key anatomical landmarks and also segment the bones (Figure 1). Comparison with other methods shows that we achieve excellent localization and segmentation accuracies. We demonstrate that the joint learning aids in achieving better performance as compared to a dedicated network for each task.

2 Method

The network architecture of the proposed method is shown in Figure 2. The network consists of a shared encoder for both the tasks with four level of down sampled convolutional blocks. At each level we have two 3D convolutional blocks with kernel size $3 \times 3 \times 3$ followed by batch normalization and leaky ReLU activation layers. We use residual learning, which helps in capturing the features more effectively.

Segmentation: The decoder on the segmentation branch consists of 4 levels of upsampling and deconvolution blocks. The features from the encoder blocks are concatenated to the corresponding levels in the decoder similar to Unet architecture. Let S_j be the segmentation mask for j^{th} segment. For K segmentation labels, K 3D volumes are created such that each of the volume has labels corresponding to one segment. The foreground is indicated by 1 and rest of the voxels are set to 0. Thus the target for the segmentation branch is a 4-dimensional array of size $W \times H \times D \times K$.

Landmark detection: Let L_i be the i^{th} landmark with $i \in \{1, 2, \ldots N\}$, where N being the number of landmarks to be detected. For the landmark L_i, we create a heatmap $h_i \in \mathbb{R}^d$, which can be interpreted as a probability distribution of a given landmark. Assuming the distribution of the landmark to follow a Gaussian distribution, we create a heatmap such that the value at any point is given by the Gaussian kernel centered at that location of the landmark. The heatmaps are scaled so that the intensity is in the range of 0 and 1. The target for the landmark detection branch is a 4-dimensional array of size $W \times H \times D \times N$, where N is the number of landmarks.

The decoder on the landmark detection branch consists of two sub-networks: a local-appearance network; and a global-context network [9]. These two sub-networks interact with each other through a point-wise multiplication ($\otimes$) as shown in the architecture. If $h_i^{LA}(\mathbf{x})$ is the output of local-appearance network and $h_i^{GC}(\mathbf{x})$ is the output of global-context network, then the final response of the network is given by:

$$h_i(\mathbf{x}) = h_i^{LA}(\mathbf{x}) \otimes h_i^{GC}(\mathbf{x}). \tag{1}$$

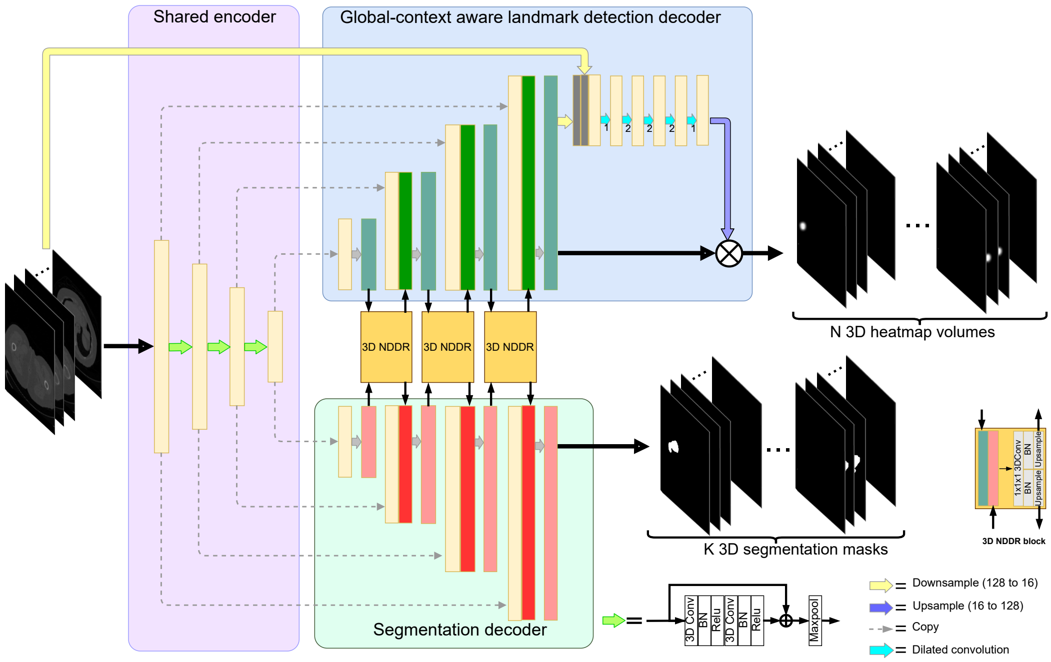


Figure 2: Proposed network architecture: consists of a shared encoder and two separate decoders; one for each task. These blocks share features via a 3D neural discriminative dimensionality reduction (3D NDDR) block. The landmark detection branch has a additional sub-network to capture the global context in which the landmark appears.

Because of the tasks getting split, the optimization process forces local-appearance network which operates at higher resolution to use all its capacity to learn locally accurate prediction but may produce ambiguous output, while leaving the task of figuring out the global information to the global-context network. The global context network operates at a much lower resolution, thus making it easier to accumulating information from the entire image space and thus gathering the context. The global context network filters out the undesirable outputs from the local appearance network through point-wise multiplication.

The local appearance network is realized using an architecture similar to the decoder of the segmentation branch described earlier. To realize the global-context network, we use a fully convolutional network consisting of five convolutional layers with different dilation factors as shown in Figure 2, with each layer followed by batch normalization and leaky ReLU activation layer. This sub-network is designed to operate at a resolution of $16 \times 16 \times 16$ and the parameters are set such that the receptive field of the sub-network is equal to this resolution. Similar to [9], in order to learn the global context better, we add a skip connection from input layer to the global-context network *i.e.,* the input layer is down-scaled and concatenated to the down-scaled output of the local-appearance network before passing it to the global context network. This helps the network to understand the context, not just from the output of local appearance, but also from the intensity variation in the input image. The output is upsampled again to the original resolution before performing the point-wise multiplication.

Cross talk feature fusion: In order to fully utilize the advantage of multi-task learning, it is well known that there has to be a mechanism of cross talk between the features of different branches. Here, we utilize the hierarchical features from the decoder branches of both the tasks. In this work, we extend the neural discriminative dimensionality reduction (NDDR) proposed in [4] to 3D and use it as the feature fusion layer. Gao *et. al.*, [4] show that discriminative dimensionality reduction aids in improved performance. To realize the 3D NDDR block, we first concatenate the task-specific features with same spatial resolution, in our case the features from same levels of decoders, and then use a $1 \times 1 \times 1$ convolution to learn the discriminative feature embedding. For simplicity we use Xavier initialization for the weight matrix.

Loss function: For the landmark detection branch, we use a mean square loss. Given the ground truth heatmap h_i for i^{th} landmark, and predicted heatmap $\hat{h}_i$, the loss of the landmark detection branch is given by $\mathcal{L}_{\text{lm}} = \text{mse}(h_i, \hat{h}_i)$. The Dice score coefficient is used as a measure to assess segmentation performance when the ground truth is available. It has also been adapted as a loss function [11] to train neural networks for semantic segmentation tasks. For the segmentation branch, we use Dice loss $\mathcal{L}_{\text{seg}}$. We combine these two loss terms and the total loss for the network is given as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{lm}} + \lambda \mathcal{L}_{\text{seg}},$$

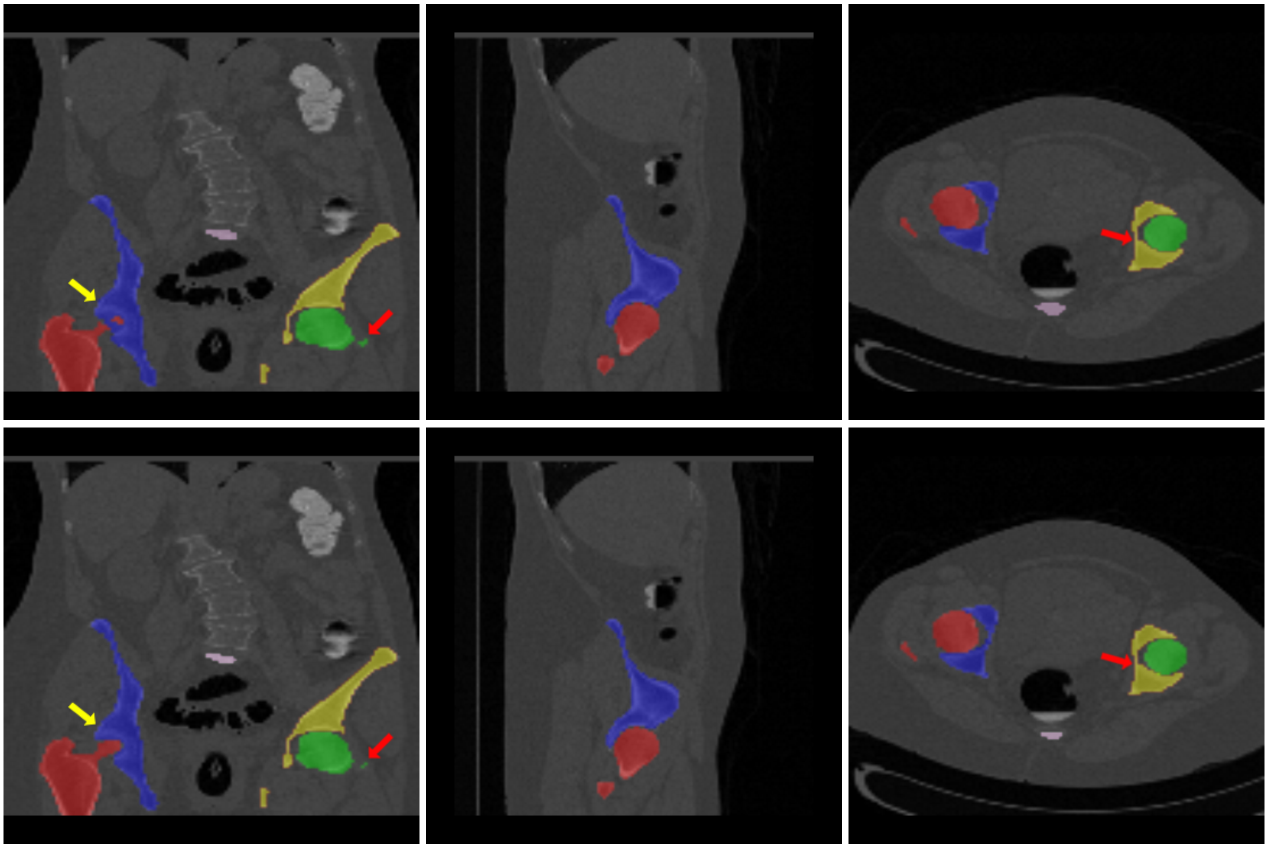


Figure 3: Segmentation Results. Groundtruth annotations (top row) and predicted results for proposed method (bottom row) overlayed on the input image. The network is able to predict very small segments with good accuracy (red arrows). In some rare cases, we observed that the network wrongly predicted the labels (yellow arrow)

where, λ is a regularization parameter that is used to balance the two loss terms so that both the tasks are given equal priority while training.

3 Experimental setup

Data: In this study, we use the abdominal CT data set provided by [12]. The dataset consists of CT scans from 90 patients. Each axial slice is of resolution 512×512. The scans are selected such that the entire pelvis region is covered in the scan. The ground truth annotation were marked by an experience radiologist and the segmentation masks for right femur, left femur, right hip, left hip and sacrum are provided. The annotation for 15 important anatomical landmarks in this region are available.

Evaluation metrics: The performance for the segmentation task is evaluated based on the Dice coefficient. For the landmark detection, the evaluation is done based on successful detection rates (SDR) and mean radial error (MRE). SDR is defined as the percentage of landmarks detected within a specified distance from the ground truth and MRE is defined as $MRE = 1/n \sum_{i=1}^{n} R_i$, where R_i is the Euclidean distance between the actual and predicted location in millimeters, and n is the number of test images.

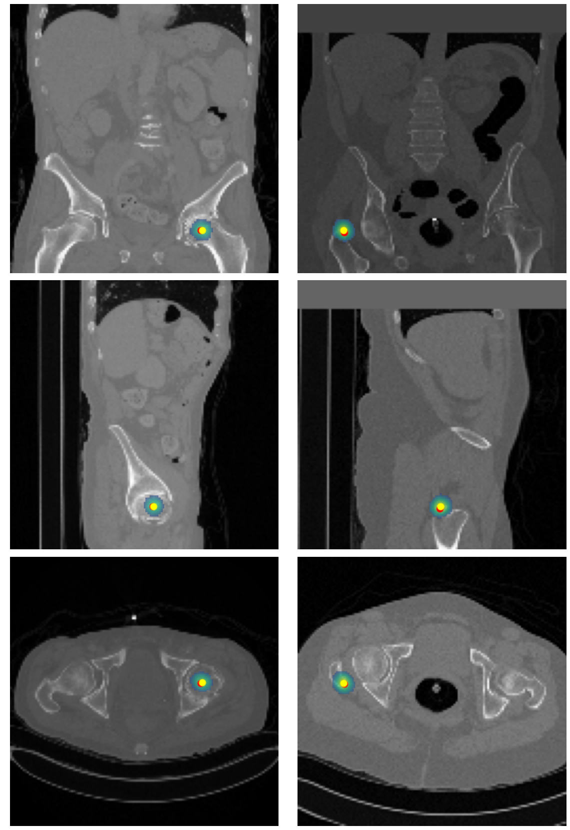
Parameter tuning and experimental setup: While training, we use the ADAM optimizer with learning rate of 10^{-4}. The input volumes are augmented with random rotations of $\pm 2^{\circ}$, translation of ± 10 pixels along all three directions, and scaling between 0.9 to 1.1 times the original size. The same transformation is applied to all of the heatmaps and segmentation labels. For the landmark detection, we set the standard deviation for Gaussian kernel as $\sigma = 11$. Through empirical evaluation, we set the regularization parameter λ in the loss function as 0.001. Since a test set is not provided with the dataset, we evaluate the results based on 5-fold cross-validation. Among the 90 images, 72 images are randomly selected and are used for training and the rest 18 are used for testing. The final evaluation metrics are provided by averaging the results from the 5-fold cross validation. The training is performed on NVIDIA V100 GPU with 32GB of RAM, which took around 15 minutes for each epoch.

Comparison methods: We evaluate our results against state-of-the-art single task methods *i.e.,* the networks which are trained specifically for either landmark detection or segmentation task. We also evaluate against a baseline multi-task network without the addition of global context network and 3D NDDR layers (referred to as *multi-task baseline*). For segmentation, we compare against Wang *et. al.* [12], which uses a network based on Unet architecture but uses both the 3D CT image and estimated volumetric shape models of the targeted bone structures as input. In Wang *et. al.* [12] the results are provided at a downscaled resolution of 128×128. For fair comparison, we run our experiments at the same resolution.

4 Results and discussion

Segmentation: Table 1 shows the comparison of performance of different methods for the segmentation task. The 3D Unet performs better than the 2D Unet trained on the individual slices emphasizing the importance of spatial context information brought in by the additional dimension. With the addition of statistical shape context, Wang *et. al.* [12] show a better performance compared to vanilla 3D Unet model. The advantage of multi-task learning can be seen from the results. The baseline multi-task model shows significant improvement compared to the baseline single-task networks. We can observe a further improvement with proposed multi-task network which outperforms all the other methods. Not only are the mean values lower, we note that the standard deviation is also significantly lower indicating higher consistency in prediction. We have also compared the results against nnU-Net [5] and slightly higher dice coefficient of 0.971 was achieved compared to the proposed method. Although it used a Unet architecture, the higher performance can be attributed to the heuristic nature of the method. It is worth noting that such an addition on top of improved architecture such as ours should further improve the results. Figure 3 shows the ground truth annotations and predicted masks for three orthogonal views. We can observe that the predicted mask is close to the ground truth. The network is able to predict very small segments with good accuracy (red arrows). In some rare cases, we observed that the network wrongly predicted the labels (yellow arrow).

Landmark detection: Figure 4 shows the output for one of the test images. The predicted heatmap for one of the landmarks (greater trochanter of right femur) is overlayed on the input image. The predicted and actual ground truth locations are also shown which indicate the predicted landmark lies very close to the ground truth. The comparison of results for landmark detection for different methods are shown in Table 2. The baseline multi-task network (*MTL baseline*) shows slight improvement over the vanilla 3D Unet. The addition of global context network to the Unet (*Unet-GC*) also shows good improvement in results. We also performed the experiment to understand the significance of global context network in the multi-task learning. As expected, it was found that the inclusion of global context network improved the results (*MTL-GC*). The proposed method shows a significant improvement and out performs other methods at all levels. From the results, it appears that the landmark detection benefits more when compared to the segmentation task with the multi-task learning framework. This can be attributed to the fact that the landmarks in this case appear on or within the boundaries of the bones which are segmented. The inclusion of global context sub-network and feature fusion using 3D NDDR aids the network to predict more accurately. The SDR at 6mm was close to 100% for all the landmarks except for apex of coccyx which was around 61%. This could be due to the fact that the coccyx, even among normal population is slightly different in each individual. Also, no specific significance is attached to the coccyx as far as



(a) (b)

Figure 4: Predicted heat map overlayed on the input image for all three views (head of left femur (a), and greater trochanter of right femur (b)). The ground truth annotation is indicated by red point and the predicted landmark by yellow.

surgical planning is concerned, hence the impact on downstream tasks is minimal. We note that the lower performance across all methods is due to the lower resolution of input and output labels. Down-sampling to $128 \times 128 \times 128$ size results in voxel resolution of roughly around 3mm across all dimensions and any small error at lower resolution has larger impact on final prediction results.

Figure 5 shows the response of various stages of landmark detection branch. The top row shows the response of the network for a true prediction for lesser trochanter of left femur (channel number 4 in our network). The output of local appearance network for the same channel shows another significant peak as shown in the bottom row. This could be because the local features around this point appears similar to the actual predicted point. The local appearance network produces locally accurate predictions but may have ambiguous peaks. However, due to the point-wise multiplication with the output of global context network, this peak is suppressed at the final output.

5 CONCLUSIONS

In this work, we propose a novel 3D multi-task network to simultaneously predict the anatomical landmarks and segmentation. The

Table 1: Comparison of segmentation results in terms of Dice coefficient (mean±std)

Method	2D Unet	3D Unet	Wang *et. al.* [12]	MTL baseline	Proposed
Right Femur	0.939 ± 0.122	0.953 ± 0.019	0.962 ± 0.018	0.971 ± 0.005	0.972 ± 0.006
Left Femur	0.925 ± 0.178	0.949 ± 0.039	0.958 ± 0.031	0.966 ± 0.008	0.974 ± 0.004
Right Hip	0.943 ± 0.054	0.947 ± 0.016	0.957 ± 0.011	0.966 ± 0.008	0.971 ± 0.005
Left Hip	0.940 ± 0.079	0.947 ± 0.016	0.958 ± 0.013	0.964 ± 0.010	0.971 ± 0.005
Sacrum	0.894 ± 0.056	0.905 ± 0.032	0.924 ± 0.027	0.940 ± 0.020	0.952 ± 0.014
Average	0.927 ± 0.098	0.940 ± 0.024	0.952 ± 0.020	0.961 ± 0.010	0.968 ± 0.005

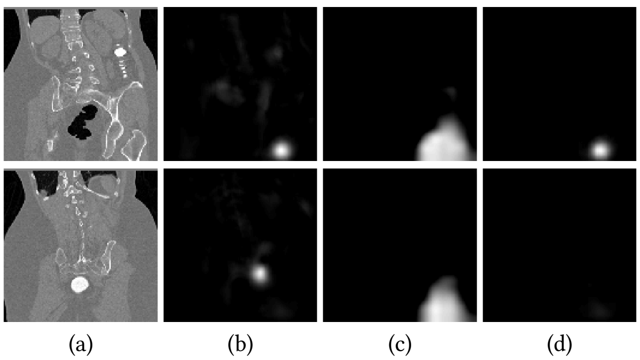


Figure 5: Output at different stages of landmark detection branch: input (a), local appearance network output (b), global context network output (c) and predicted heatmap (d). (Top row) True prediction for one of the landmarks (lesser trochanter of left femur). (Bottom row) Local appearance network shows a peak for the same channel, but is filtered out by global context network.

Table 2: Performance comparison of landmark detection

Method	MRE (mm)	SDR(%)			
		2mm	3mm	4mm	6mm
3D Unet	5.67 ± 3.008	3.70	14.07	35.18	62.23
Unet-GC	4.56 ± 2.840	11.85	29.25	52.96	78.14
MTL baseline	5.18 ± 3.178	9.25	22.96	45.56	68.89
MTL-GC	4.18 ± 2.818	12.22	40.37	57.40	84.07
Proposed	2.88 ± 1.670	31.12	63.36	80.00	95.19

end-to-end trainable network uses both local and global information for accurate predictions. The feature fusion *via* discriminative dimensionality reduction allows the network to share the features among these highly related tasks. The method was applied on pelvic CT data, and excellent results were demonstrated for both the tasks when compared with networks trained for single task. Future work includes the use of these predictions in downstream tasks such as digital templating and implant size selection.

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